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How does practice change across the season? A descriptive study of the training structures and practice activities implemented by a professional Australian football team

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Abstract

There is limited research investigating objective practice monitoring in team sports. This observational study examined the practice activities used by eight professional coaches across 72 different practice sessions in one season within a professional Australian football team. This study aimed to evaluate the extent to which these practice types differed from those shown to facilitate skill acquisition in team sports. Across the whole season (pre-season and in-season), coaches implemented seven practice types which were categorised into training-form and playing-form. Practice type frequencies were different between pre- and in-season ($\chi^2 = 109.6$, $p < 0.001$). Significant differences were reported for practice type duration between pre-season and in-season ($p = 0.023$, $d = -0.17$), where in-season activities were longer. Coaches implemented a higher percentage of practice time to playing-form activities in-season (62.0 ± 25.4) compared to pre-season (52.4 ± 19.8) [95%CI: –2.35:21.4]. Coaches were more likely to alter the frequency of practice types, rather than the exposure time. Players performed more training-form activities during pre-season than in-season ($\chi^2 = 30.8$, $p < 0.001$). There were no differences in playing-form activities between pre- and in-season. This study provides insights into the practice activities implemented by professional Australian football coaches and discusses the extent to which they represent best practice in the context of skill acquisition.

Keywords

Constraints-led approach, representative learning design, skill acquisition, training

Introduction

Experts exhibit superior execution, perceptual-motor proficiency and information processing abilities with respect to skill in team sports.[1,2] The amount of time spent in practice has historically been associated with the above characteristics e.g. Ericsson and colleagues[3] hypothesised improvements in skill were primarily related to the amount of time spent in deliberate practice.[3] However, contemporary research suggests that the development of skilled performance is more complex. Recent research has demonstrated a range of constraints that impact skill development and proficiency. For example, although Baker and Cobley[4] note individual proficiency when performing a skill may originate from the relationship between the amount of practice and performance, it is the type of practice

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(i.e. high validity environments and the information available within them) that can further differentiate expert performers from non-experts.[5,6] Different practice environments in team sport lead to different behaviours emerging.[7] Thus, the consequence of the interaction of the constraints imposed on the skill is fundamental to how skill emerges and develops.[8] Consequently, an awareness of the specific environments implemented for skill acquisition and expertise development is useful for practice design. In part, this will enhance coach's awareness for skill acquisition and development and its usefulness for future refinement of practice design.

According to the constraints-led approach to skill acquisition, skilled behaviour emerges through the interaction of individual, environmental and task constraints.[9,10] Due to the complexity of team sport performance, performers interact and adapt to an array of interacting constraints to collectively achieve a common goal.[9,11] Consequently, coaches aim to design practice that represents the performance contexts and the information sources that are present during competition.[12] Task constraints are commonly implemented by coaches[13] and involve changing the field size, the number of players, or task goals, to name a few. Coaches often manipulate task constraints during practice to elicit specific behaviours from their performers. Nonetheless, these constraints may not always be manipulated deliberately, and in reality are often the result of ad-hoc implementations by coaches (e.g. the players are not sufficiently pressured, so the surface area in a small-sided game is adapted during training). It is this exposure to specific practice contexts that allows performers to adapt their behaviour to the environmental information more effectively over time. This concept is referred to as perceptual attunement and is commonly associated with expertise in human movement.[14,15] This mutuality between the performer and the performance environment is adequately described in the ecological dynamics approach to skill acquisition in team sport.[15,16] To effectively train collective and individual behaviour in team sports, exposure to practice contexts that replicate the coupling of perception and action is needed. Interestingly, the specific practice activities implemented by coaches throughout the season at the professional or amateur level has yet to be documented in the scientific literature.

A common approach within team sports training is to increase specificity as competition approaches.[17,18] In the Australian Football League (AFL), particular attention is payed to periodisation and the specific activities players perform across pre-season and in-season. For example, Moreira and colleagues[19] reported higher physical training loads for all training activities during the pre-season compared to in-season. The emphasis for practitioners is to physically 'peak' the players for the competitive in-season period, which is commonplace amongst physical performance programs in the AFL.[19,20] Consequently, practitioners and coaches must simultaneously prepare players physically, but also skilfully and tactically. The subsequent structure of practice and how much time is spent practicing certain activities then becomes a considerable concern. Importantly, when and how this shift in practice occurs often demonstrates a different collective focus for teams and likely reveals the value coaches place on physical, skill or tactical development throughout the season. Guided by physiology periodisation, Farrow and Robertson[21] and Otte and colleagues[22] provided conceptual frameworks for skill acquisition development in team sport and skill development for 'specialist' coaches, respectively. Both frameworks consider traditional coaching features that can be observed to indicate skill acquisition periodisation, such as the progression of practice (e.g. less to more representative practice), incorporating more specific practice closer to competition and how to overload the organisation of the team and individuals. Whilst these frameworks propose guidelines for skill acquisition periodisation, no literature exists to determine whether coaches implement such guidelines. It is generally coaches relying on their experience, expertise and philosophies when designing and implementing practice. Hence, it would be beneficial to notate and measure the types and duration of practice coaches implement to help enhance our understanding of how much value coaches place on specific types of practice and when this changes throughout the season. There is currently an absence of research illustrating the types of practices professional teams perform in a full season across any sport.

The AFL is the professional competition of Australian Football (AF) that involves 18 competing teams based around Australia. The complex nature is demonstrated through dynamic environments and emergent patterns of behaviour.[23,24] Moreover, coordinative team behaviour in AF is predicated on functional behaviours based on the spatiotemporal relations between performers, as they continuously adapt to their teammates and the opposition's movements.[25] This exemplifies that AF players perceptually attune to relevant sources of information in games and take advantage of shared affordances, i.e. collective opportunities for action[7] to produce actions that increase the team's likelihood of winning. As such, representative sources of information and action should be reflected in practice contexts through representative learning design.[12] Recent research in professional AF highlights the discord between the constraints encountered in training environments and competitive matches.

These studies provide clarity for practitioners aiding practice design. For example, Ireland and colleagues[26] described conflicting temporal and situational constraints when disposing possession in competitive matches compared to practice. Specifically, defensive pressure applied to kicking disposals was significantly higher in matches compared to practice contexts. Similarly, Browne and colleagues[27] compared the frequency and interaction of constraints across match simulation, small-sided games and competitive matches. The authors reported disparity between each context, especially in regard to the disposal type i.e. kick or handball, and the proceeding disposal effectiveness. Here, the sequence of an effective third disposal was lower in matches and match simulations compared to the observed small-sided games. Thus, understanding and implementing representative learning design is crucial and can assist coaches to provide practice environments that are more likely to increase the transfer of practice to game performance.[6,12]

To date, no studies have documented the practice activities that professional AFL teams perform throughout a complete season. This study aimed to describe the practice activities implemented by AF coaches throughout a competitive season, with an emphasis on quantifying the exposure (time and frequency) to certain types of practice. It was hypothesised that coaches would implement a higher proportion of more representative activities, termed playing-form activities, during the in-season phase. Additionally, it was hypothesised that coaches would employ a higher proportion of less representative activities, termed training-form practices, in the pre-season compared to in-season.

Materials and methods

Participants

One professional AF club agreed to be involved in the research. Specifically, the coaching group and head of performance agreed for this research to be executed. The team consisted of 48 male professional Australian footballers (age: 23.28 ± 4.35 years) and they participated in their normal training across one entire season. The pre-season phase lasted four months from November to March, whilst the in-season phase lasted 6 months from March to September. The eight senior and development coaches within the professional club had an average of 6.8 ± 6.5 years of AFL coaching experience, in addition to an average of 11.5 ± 5.1 years of playing professional level AF. Training occurred without intervention from researchers. Ethics approval was obtained for this study from the University's ethics committee.

Practice activities

In total, 37 pre-season and 35 in-season training sessions were recorded across the data collection period, resulting in an entire sample of 72 sessions. The data for each training session was collected at the team level. Practice sessions were planned by the coaching group in conjunction with physical performance staff to determine player availability (which players can participate in which drills). The club utilises a long-term planning framework to guide practice design, while also accommodating for variation based on player availability e.g. if less players were available than expected, then an activity tended to be conducted with slightly smaller field dimensions. For each field session, details of practice activities were recorded and activities were filmed. Filming was captured using a combination of handheld and remote video cameras (Axis Q6128-E Network, Axis, Lund, Sweden and Canon Legria HFG30, Canon, Tokyo, Japan). One was placed perpendicular to the direction of play at the side-line, and the other behind the goals. Both cameras were placed at elevated positions. Training footage was manually coded for time spent in each practice and one investigator notated the start and finish of each practice.

All practice types were categorised into one of seven classifications by the coaching staff: fundamentals, fitness, skills, breakdown, transition, line specific and match simulation (Table 1). These practices were classified by the research team as either training-form or playing-form, as per the methods of Ford and colleagues.[28] Training-form practice types were performed in isolation or in small groups. They generally involved simplified practices[29] and are repetitive in nature. Playing-form practice categories have a game-related focus and emphasise certain phases in a game (i.e. stoppages) or relate to transitioning between certain phases (e.g. attack to defence). This practice includes position-specific set ups and break downs. All practices sessions were annotated by the same experienced observer, who had annotated more than 200 h of AF training and match-play. Coding reliability was assessed using an intra-class correlation coefficient (3, k).[30]

Data analysis

The time spent in each practice activity type was calculated over the whole season, and separately for pre-season and in-season phases. Cross-tabulations and chi-square tests of independence (χ^2) were used to investigate the relationship between practice categorisations (practice types and practice forms), and season phases (pre- and in-season). Practice type proportions were expressed as a percentage of total time spent in

Table 1. Training activity categories and their definitions.

Practice activity	Definition
Training-form	
Fundamentals	Drills that focus on kicking, handballing, marking, tackling technique and handwork Generally performed in small groups that are repetitive and blocked and have unopposed skill repetition
Fitness	Activities aimed at improving game-related fitness qualities (e.g. aerobic capacity, repeat speed ability etc.)
Skills	Consists of various practices They largely involve ball movement (cone to cone), chaotic handball practices with outnumbered situations (e.g. 4v3), goal kicking and crumbing Tend to re-enact isolated game incidents
Playing-form	
Breakdown	Part-practice activities that involve a certain phase of game play E.g. stoppages or exiting the defensive 50 Generally, these practices have limited/reduced opposition
Transition	Activities such as small-sided games that aim to simulate the continual adaptation between attack, contested play and defence seen in match-play
Line-specific	Practices that are run by each line coach and are positional and craft specific For example, midfielders may work on the centre bounce clearance whereas defenders and forwards may work on 1 on 1 marking contests
Match Simulation	Involves representative game play/simulation This involved anywhere between 13–18 players on each team

practice by dividing the time spent in the activity by the total time spent in practice for the whole season and then for pre- and in-season individually, and then multiplying this number by 100. The time in each practice activity type between pre-season and in-season was compared using a paired samples t-test. Additionally, comparing the duration of time spent in playing-form activities from pre-season to in-season was analysed using a paired samples t-test. The data was normalised by calculating the percentage of session duration players spent in playing-form activities to prevent violating the statistical assumption of independence. Similar methods normalising playing-form activities have been used previously.[28,31] Effect size statistics were used to quantify the magnitude of difference between the groups, with interpretation of magnitudes being $<0.19 =$ trivial, 0.2–0.49 = small, 0.5–0.79 = moderate, $>0.8 =$ large.[32] The alpha level for significance was set at $p < 0.05$. All statistical analyses were conducted using JASP Team.[33]

Results

The intra-class correlation coefficients between test 1 and test 2 demonstrated excellent reliability ($r = 0.97$–0.99) for the annotations of practice types and time spent in each practice type.

Practice type and practice form frequencies

Table 2 demonstrates the chi-square contributions for each practice type and practice activity form. The chi-square test of independence comparing in-season practice type frequencies to pre-season practice type frequencies demonstrated significantly different frequency distributions in different phases of the season ($\chi^2 = 109.6$, $p < 0.001$). Table 2 presents the observed and expected counts for each practice type and shows the contribution to the final chi-squared value. The large chi-square value is largely caused by not performing any fitness sessions in-season, moving to a very low amount of fundamentals in-season and an increased number of line specific practice drills in-season. Pre-season included more breakdown (+8), fitness (+63), fundamentals (+105), skill (+14) and transition (+10) type practice. Higher amounts of line specific (+14) and match simulation (+4) practice types were performed in-season. Furthermore, the chi-square test of independence demonstrated a significant difference between training-form and playing-form activities when comparing pre- and in-season frequencies ($\chi^2 = 30.8$, $p < 0.001$). Table 2 presents the observed and expected counts for each practice form and their respective contribution to the final chi-squared value. There was no change in the frequency of playing-form activities between pre- and in-season. However, the relative contribution to chi-square values is caused by a large difference in the frequency of training-form practice between pre- and in-season. Training-form practice was performed more frequently during pre-season (+182). Playing-form practice frequency was equal between pre- and in-season (82).

Practice activity duration

The average practice activity time for a drill in pre-season was 7.0 ± 0.8 mins. While the average practice activity time in-season was 8.0 ± 0.4 mins. There was a significant difference between practice activity duration between pre-season and in-season ($t(177) = -2.286$, $p = 0.023$, $d = -0.17$ [95% CI: –0.319 to –0.023]).

Table 2. Cross tabulations and chi-square contributions for each practice type and practice form.

Practice type and form	Season phase	Observed counts	Expected counts	Sums	Squared difference (Observed-Expected)	Chi squared contribution
Breakdown	Pre-season	23	255	38	6	024
	In-season	15	125			048
Fitness	Pre-season	63	422	63	4328	1026
	In-season	0	208			2080
Fundamentals	Pre-season	117	864	129	9363	1084
	In-season	12	426			2198
Line specific	Pre-season	13	268	40	1902	710
	In-season	27	132			1440
Match simulation	Pre-season	19	281	42	8	296
	In-season	23	139			601
Skill	Pre-season	97	1206	180	5551	460
	In-season	83	594			934
Transition	Pre-season	27	295	44	61	021
	In-season	17	145			042
					Total χ^2	1096
Training-form	Pre-season	277	2492	372	7752	31
	In-season	95	1228			63
Playing-form	Pre-season	82	1098	164	7752	71
	In-season	82	542			143
					Total χ^2	308

χ^2=chi-square value.

Subsequently, the in-season phase had longer practice activity durations with lower variation.

Time-use analyses

Table 3 outlines the time spent in each practice type separated by season phase. Figure 1 illustrates the proportion of time and the proportion of frequencies in each practice type between pre- and in-season. Since practice activity time differed across pre-season and in-season, a paired samples t-test was used to determine the difference in percentage of session duration spent in playing-form activities. The average percentage of time spent in playing-form activities for pre-season was 52.4 ± 19.8, whilst for in-season it was 62.0 ± 25.4. The difference in proportions between pre-season and in-season was insignificant ($t = (27)$ –1.641, $p = 0.112$, d=–0.31 [95% CI: –0.687:0.072). Across the whole season, a higher proportion of time was spent in training-form activities compared to playing-form activities. In pre-season, a higher proportion of training time was spent in training-form compared to playing-form activities, whilst a higher amount of time was spent in playing-form activities in-season, compared to training-form activities.

Across the whole season, the proportion of time in training was the largest for skill, followed by match simulation, fundamentals, transition, line specific, fitness and breakdown. In pre-season, skill type practices composed the largest proportion of training, followed by fundamentals, fitness, transition, breakdown, match simulation and line specific. The greatest proportion of training in-season was spent in skill, followed by match simulation, line specific, transition, breakdown and fundamentals. All of the relevant proportions (in percentages) are reported in Table 3.

Discussion

This descriptive study examined the practice contexts that coaches implemented in a professional AF team with a view to facilitating optimal skill acquisition principles. Practice type frequencies between pre-season and in-season were different. The difference largely originated from no specific fitness drills being performed in-season, a low amount of fundamental practice types being performed in-season and an increase in line specific practice in-season. Moreover, coaches more frequently implemented training-form practice during pre-season than in-season. The significant chi-square indicates larger contributions from the observed playing-form frequencies compared to what was expected for both in-season and pre-season. There were no differences detected between in-season and pre-season for playing-form practice frequency (82 each). There was a significant difference between practice type duration between pre-season and in-season, with drills typically being of longer duration

Table 3. Practice types and the average and total time and subsequent proportion spent in each, according to whole season, pre-season and in-season.

Practice type	Average duration (mins)	N	Total time in practice type (mins)
Whole season			
Training-form	**5.9 ± 2.8**	**372**	**2207**
Playing-form	**12.9 ± 5.1**	**164**	**2114**
Breakdown	10.9 ± 5.5	38	414
Fitness	6.8 ± 4.0	63	431
Fundamentals	4.4 ± 2.1	129	569
Line Specific	12.1 ± 4.0	40	485
Match Simulation	15.8 ± 3.4	42	664
Skill	6.7 ± 2.2	180	1207
Transition	12.2 ± 5.7	44	551
Pre-season			
Training-form	**5.8 ± 3.0**	**277**	**1609**
Playing-form	**14.8 ± 4.6**	**82**	**1220**
Breakdown	13.5 ± 4.9	23	310
Fitness	6.8 ± 4.0	63	431
Fundamentals	4.2 ± 2.1	117	495
Line Specific	15.0 ± 4.0	13	194
Match Simulation	15.8 ± 4.2	19	301
Skill	7.0 ± 2.3	97	683
Transition	15.0 ± 5.1	27	415
In-season			
Training-form	**6.3 ± 2.0**	**95**	**598**
Playing-form	**10.9 ± 4.7**	**82**	**894**
Breakdown	7.0 ± 3.9	15	104
Fitness	0 ± 0	0	0
Fundamentals	6.1 ± 1.4	12	74
Line Specific	10.8 ± 3.2	27	291
Match Simulation	15.8 ± 2.8	23	363
Skill	6.3 ± 2.0	83	524
Transition	8.0 ± 3.7	17	136

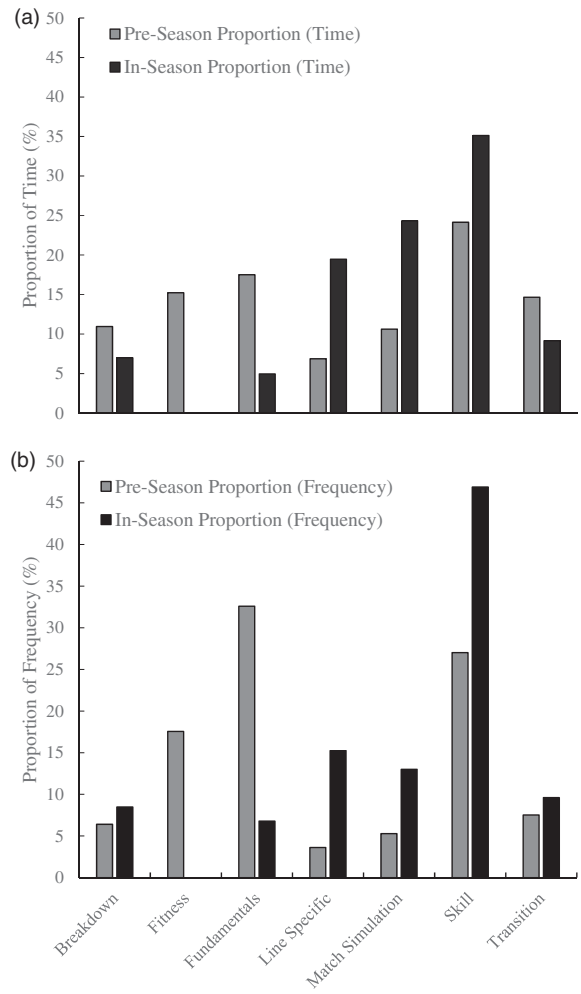

Figure 1. Proportion of practice types for time (a) and frequency (b) between pre- and in-season.

in-season. Furthermore, the mean percentage of time spent in playing-form in-season (62.0 ± 25.4) did not significantly differ compared to pre-season (52.4 ± 19.8). Collectively, these results suggest that coaches prefer to implement higher exposures, via frequency, to more practice types rather than a longer exposure to a specific, and smaller amount of practice types. For example, within one training week the coaches may implement 12 different variations of these practice classifications, with each lasting 6 min in duration. This would be in contrast to the coaches implementing 6 practice drills lasting 12 min each. The variation and differences within each practice type between pre- and in-season could also reflect variations between the intensity of these training activities. There are several conceivable reasons for these outcomes including the requirement to meet the physical requirements determined by physical performance staff, the implicit provision of more opportunities to expose players to the same problem in different contexts, or providing the coaches a platform to observe, refine and correct performance more often.

The highest pre-season proportions of training were skill (24.2%), fundamentals (17.5%) and fitness (15.2%) practice types. In contrast, in-season training time proportions were skill (35.1%), match simulation (24.3%) and line specific (19.5%) practice types. These results clearly demonstrate that coaches replace fundamentals and fitness with match simulations as they move to in-season training. This alludes to the specificity of practice hypothesis.[34] Coaches seem to implement more representative practice with reduced available time for training in-season. The match simulation may be implemented as it provides players with the ability to use more appropriate information from the environment to support actions,[12] showcasing to coaches whether the collective and individual perceptual and motor processes are desirable and align with

specific tactics and outcomes they are wanting to observe. From an ecological view, it aims to create shared affordances and for the players to become attuned to these shared affordances that would also be present in competitive matches. The collective and cyclical relationship between shared affordances and shared action exposes players to behaviours, constraints and information that can be observed in match-play.[7] Hence, match simulation may provide the most representative and manipulatable degree of perception-action coupling that can seemingly be achieved in practice. Specific constraints, like space (m^2) per player, using a scoreboard and assigning specific criteria to be met and achieved during match simulation, can be manipulated to optimise collective adaptations during practice.

Skill type practice was the most frequently performed in both the pre- and in-season phases. Moreover, the tendency for an increase of training time proportion performing skill type practice in-season compared to pre-season (+10.9%) indicates that coaches appear to implement a strategy that changes from general to more specific training throughout the season. This aligns with skill periodisation ideas proposed previously.[21,22] Interestingly, in-situ skill periodisation is not as complicated as that proposed in the models and is largely implemented ad-hoc. Nonetheless, the large proportion of time spent performing skill type activities suggests coaches are primarily concerned with improving their athletes' skill proficiency. According to the literature, this approach to training may not be the best approach to achieve that goal.[12,34] This practice type largely involved isolated game incidents that removed a continuation of play. Accordingly, this practice type may remove key perception and action responses for subsequent phases of play. For skill to improve and transfer to competitive match-play to occur, high validity environments, where performer-environment interactions represent those experienced during game play, are needed.[12] Here, we advocate increased opportunity for skill acquisition specialists in professional sport to play a key role in continually developing expertise. As Steel and colleagues[35] describe, "skill acquisition specialists are sport scientists who examine the theories, principles and processes of motor and perceptual learning". Thus, a complimentary role between coaches, who make decisions based on experiential knowledge, and skill acquisition specialists, who make decisions based on both empirical and experiential knowledge can lead to beneficial outcomes for practice design, feedback and instruction. Furthermore, the complimentary role skill acquisition specialists can play ensures the continual development of perceptual-motor learning based on scientific principles. A good example of how the cross-disciplinary nature of a skill acquisition specialist can help improve practice design has been documented by Ireland and colleagues.[26] These authors reported that kicking disposal constraints relating to space and time were underrepresented during practice compared to professional AF matches. For expert performers to improve skill, informational sources used to regulate behaviour need to be very specific[6] and as such, it is likely that until coaches use and develop practice activities that truly resemble conditions from match-play, skill proficiency will not be optimised, or will remain at a level below that required for successful performance in competition. A simple method to enhance skill development in experts is to perform the skill at speeds similar to game play so the performers can process relevant information and execute skills at the required pace.[36] However, rather than mimicking actual game-play pace, coaches often choose to train skilled behaviour and decision making at a proportional pace of that experienced during competition (e.g. match simulation run-throughs at 60% of maximal velocity).

Similar to previous research assessing coaching and youth practice contexts in elite association football,[1,28,31] these findings indicate an insignificant difference but higher proportion of time in-season vs pre-season is spent performing playing-form activities. Moreover, the greatest contributor to the chi-squared value (χ^2=30.8) assessing the practice form activity frequencies was the difference between expected and observed frequencies of playing form activities in-season. That is, the frequency of playing-form activities in-season was much larger than expected. This may indicate an increased and implicit awareness of representative practice from coaches, or an enhanced understanding of transferable practice due to their experience and philosophies. Additionally, these results may highlight the multidimensionality of practice design concerns for professional coaches. For example, other factors such as training load management e.g. reduce the number of defenders to reduce the number of change of direction efforts, the players and teams experience, the number of days between competitive games in-season e.g. less days between games means more time is focussed towards recovery and not training, meaning the most relevant activities must be performed, or the upcoming opposition e.g. positional requirements across different phases of play. Nonetheless, playing-form activities are deemed more relevant as they generally contain more specific informational sources such as pressure from defenders, spacing and game speed. However, these proportions differed in pre-season for the current data i.e. training-form encompassed 56.9% of total training time.

A conservative view may suggest slightly decreasing the proportion of training-form activities, whilst slightly increasing playing-form activities.[28] This may be an ambitious suggestion in practice, but this approach considers that expert performers are systematically characterised by enhanced domain-specific perceptual-cognitive abilities such as enhanced scanning and pattern recognition, anticipation and decision making[37] whilst measures of basic visual and cognitive function (i.e. domain-general skills) do not differ.[38] Thus, expert performers, as showcased here through a professional AF team, require consistent exposure to specific training to enhance adaptations related to their domain-specific expertise. Consequently, an integrated approach to increase playing-form practice types would be recommended across the whole season. Nonetheless, the overall proportion of training-form to playing-form was very similar (51.1% and 48.9%, respectively) over the whole season. Regular assessments and continued monitoring by performance analysts can provide feedback to coaches and assist their quest to optimise practice contexts.

Fundamental type practice constituted 13.2% (pre-season: 17.5%, in-season: 5%) of total training time across the whole season. These training-form practices incorporate basic variations of closed skill performances (e.g. groundballs, handballing, kicking, marking, tackling and handwork). The performance of the skill is generally completed in a decontextualized manner where skill execution is independent of the context it is required in. As such, it may not provide the specific information that experts use to regulate match behaviour and decision-making or may lack processing requirements that develop domain-specific skill.[6] This practice type also requires little cognitive attention as the athletes tend to be extremely proficient at the movement solutions. Additionally, because team sport experts differ to non-experts regarding pattern recognition, automatic movement control, anticipation and coping with constraints,[6] repeated exposures to simple closed skill performances are less representative of the domain-specific exercises required for experts to learn and keep improving. Although it can be utilised as a 'get hands on the footy' exercise, from a skill acquisition perspective for experts, it likely offers very little. For example, Farrow and colleagues[2] reported higher cognitive complexity (and thus, engagement) in open type skill contexts. Thus, closed skill practice contexts likely do not cognitively engage the players and do not challenge the players. It appears that the utility of such exercises and practice contexts for expert performers is not warranted and time could be better utilised engaging in other practice types.

Whilst this was the first study to report on practice types and contexts implemented by expert coaches in a professional AF team, some limitations must be considered. Specifically, the data was collected from one professional AF team and therefore may lack generalizability to other professional AF teams' programs or more broadly to other team sports. However, the methodology presented in this study may be useful for other researchers and practitioners to adopt and investigate their own practice environments. Additionally, as these results are clearly preliminary and rely somewhat on the descriptive information on each practice type, future research is required to strengthen the interpretation and extrapolation to more teams and indeed different sporting contexts. This will assist researchers and practitioners understand how expert performers can continue to develop and refine their performance, whilst further understanding where research efforts are required. Nonetheless, the longitudinal collection over one full season using professional athletes presented here strengthens the interpretation of the data and the ecological validity.

Conclusion

This study provided information on the training structures and practice contexts that coaches implement in a professional AF team. There were seven practice types the coaches implemented throughout the season which were different between pre- and in-season phases. Skill practice was most prominent throughout the whole season but was performed more frequently in pre-season compared to in-season. This signifies the value coaches place on developing skill proficiency in professional AF players before competitive matches begin. Coaches also tended to emphasise training-form activities in the pre-season and playing-form activities in-season. This likely reflects coaches' willingness to implement practice that transfers into completive play by providing activities with more representative learning design in-season. While these coaches' practices generally correspond with scientific knowledge surrounding optimal practice design, some training activities, including the large proportion of time spent performing closed-natured skills, and practices performed unopposed, may indicate the opportunity for skill acquisition specialists to aid in optimising practice design. Skill acquisition specialists can play an important complimentary role to help coaches design practice and provide feedback and instruction. Additionally, they can implement pedagogical frameworks across an organisation through evidence-based practice.

The information from this study can be used to monitor the type of practice being performed and used to determine the duration and exposures to certain practice types. Collectively, future research and daily practice in sport may warrant these analyses to

be used alongside the constraints-led approach and in conjunction with collective behavioural measures to determine behavioural responses to practice in Australian football.

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References

1. Williams AM and Hodges NJ. Practice, instruction and skill acquisition in soccer: challenging tradition. *J Sports Sci* 2005; 23: 637–650.
2. Farrow D, Pyne DB and Gabbett T. Skill and physiological demands of open and closed training drills in Australian football. *Int J Sports Sci Coach* 2008; 3: 485–495.
3. Ericsson KA, Krampe RT and Tesch-Römer C. The role of deliberate practice in the acquisition of expert performance. *Psychol Rev* 1993; 100: 363–406.
4. Baker J, Schorer J and Cobley S. *Talent identification and development in sport: international perspectives*. London: Routledge, 2012.
5. Kahneman D and Klein G. Conditions for intuitive expertise: a failure to disagree. *Am Psychol* 2009; 64: 515–526.
6. Farrow D, Baker J and MacMahon C. *Developing sport expertise: researchers and coaches put theory into practice*. London: Routledge, 2013.
7. Silva P, Garganta J, Araújo D, et al. Shared knowledge or shared affordances? Insights from an ecological dynamics approach to team coordination in sports. *Sports Med* 2013; 43: 765–772.
8. Newell KM. Constraints on the development of coordination. In: Wade MG and Whiting HTA (eds) *Motor development in children: aspects of coordination and control*. Dordrecht: Martinus Nijhoff, 1986, pp.341–360.
9. Davids K, Araújo D, Shuttleworth R, et al. Acquiring skill in sport: a constraints-led perspective. *Int J Comput Sci Sport* 2003; 3: 31–39.
10. Davids K, Button C and Bennett S. *Dynamics of skill acquisition: a constraints-led approach*. Champaign: Human Kinetics, 2008.
11. Handford C, Davids K, Bennett S, et al. Skill acquisition in sport: some applications of an evolving practice ecology. *J Sports Sci* 1997; 15: 621–640.
12. Pinder RA, Davids K, Renshaw I, et al. Representative learning design and functionality of research and practice in sport. *J Sport Exerc Psychol* 2011; 33: 146–155.
13. Ometto L, Vasconcellos FVA, Cunha FA, et al. How manipulating task constraints in small-sided and conditioned games shapes emergence of individual and collective tactical behaviours in football: a systematic review. *Int J Sports Sci Coach* 2018; 13: 1–15.
14. Gibson JJ. *The ecological approach to visual perception*. Boston: Houghton Mifflin, 1979.
15. Araújo D, Davids K and Hristovski R. The ecological dynamics of decision making in sport. *Psychol Sport Exerc* 2006; 7: 653–676.
16. Davids K, Araújo D, Vilar L, et al. An ecological dynamics approach to skill acquisition: implications for development of talent in sport. *Talent Dev Excel* 2013; 5: 21–34.
17. Bompa TO. *Periodization training: theory and methodology*. 4th ed. Champaign: Human Kinetics, 1999.
18. Impellizzeri FM, Rampinini E and Marcora SM. Physiological assessment of aerobic training in soccer. *J Sports Sci* 2005; 23: 583–592.
19. Moreira A, Bilsborough JC, Sullivan CJ, et al. The training periodisation of professional Australian football players during an entire AFL season. *Int J Sports Physiol Perform* 2015; 10: 566–571.
20. Robertson S and Joyce D. Evaluating strategic periodisation in team sport. *J Sports Sci* 2018; 36: 279–285.
21. Farrow D and Robertson S. Development of a skill acquisition periodisation framework for high-performance sport. *Sports Med* 2017; 47: 1043–1054.
22. Otte FW, Millar SK and Klatt S. Skill training periodisation in "specialist" sports coaching – an introduction of the "PoST" framework for skill development. *Front Sports Act Living* 2019; 1: 1–17.
23. Alexander JP, Spencer B, Mara JK, et al. Collective team behaviour of Australian rules football during phases of match play. *J Sports Sci* 2019; 37: 237–243.
24. Sheehan WB, Tribolet R, Watsford ML, et al. Using cooperative networks to analyse behaviour in professional Australian football. *J Sci Med Sport* 2020; 23: 291–296.
25. Silva P, Vilar L, Davids K, et al. Sports teams as complex adaptive systems: manipulating player numbers shapes behaviours during football small-sided games. *SpringerPlus* 2016; 5: 10.
26. Ireland D, Dawson B, Peeling P, et al. Do we train how we play? Investigating skill patterns in Australian football. *Sci Med Football* 2019; 3: 265–274.
27. Browne PR, Woods CT, Sweeting AJ, et al. Applications of a working framework for the measurement of representative learning design in Australian football. *PLoS ONE* 2020; 15: e0242336.
28. Ford PR, Yates I and Williams AM. An analysis of practice activities and instructional behaviours used by youth soccer coaches during practice: Exploring the link between science and application. *J Sports Sci* 2010; 28: 483–495.
29. Edwards WH. *Motor learning and control: from theory to practice*. Wadsworth: Cengage Learning, 2010.
30. Koo TK and Li MY. A guidelines of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med* 2016; 15: 155–163.

31. Roca A and Ford PR. Decision-making practice during coaching sessions in elite youth football across European countries. *Sci Med Football* 2020; 4: 263–268.
32. Cohen J. *Statistical power analysis for the behavioural sciences*. 2nd ed. Hillsdale: Lawrence Earlbaum Associates, 1988.
33. JASP Team 2020. JASP (Version 0131)[Computer software].
34. Proteau L. On the specificity of learning and the role of visual information for movement control. In: Proteau L and Elliott D (eds) *Vision and motor control*. Amsterdam: North Holland, 1992, pp.67–103.
35. Steel KA, Harris B, Baxter D, et al. Coaches, athletes, skill acquisition specialists: a case of misrecognition. *Int J Sports Sci Coach* 2014; 9: 376–378.
36. Reimer T, Park ES and Hinsz VB. Shared and coordinated cognition in competitive and dynamic task environments: an information-processing perspective for team sports. *Int J Sport Exerc Psychol* 2006; 4: 376–400.
37. Williams AM, Fawver B and Hodges NJ. Using the 'Expert Performance Approach' as a framework for examining and enhancing skill learning: Improving understanding of how experts learn. *Frontline Learn Res* 2017; 5: 139–154.
38. Voss MW, Kramer AF, Basak C, et al. Are expert athletes' 'expert' in the cognitive laboratory? A meta-analytic review of cognition and sport expertise. *Appl Cognit Psychol* 2010; 24: 812–826.

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